

Build a Data Mart in SQL

Phase 1 – Conception Phase

Course: DLBDSPBDM01

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Project Overview (Adjusted After Tutor Feedback)

1. Requirements Analysis

The design process began with a thorough analysis of the platform's requirements, focusing on identifying the main roles—Guests, Hosts, and TravelAdmins—and the actions each of these roles perform. This version has been updated in line with tutor feedback.

Roles and Actions:

Guests: Guests interact with the platform by searching for properties, booking rentals, leaving reviews, and managing their profiles. The database must support these activities by storing and retrieving the necessary data.

Hosts: Hosts are responsible for listing properties, managing bookings, and responding to guest inquiries, and leaving reviews. The database tracks property details, availability, and the actions of each host.

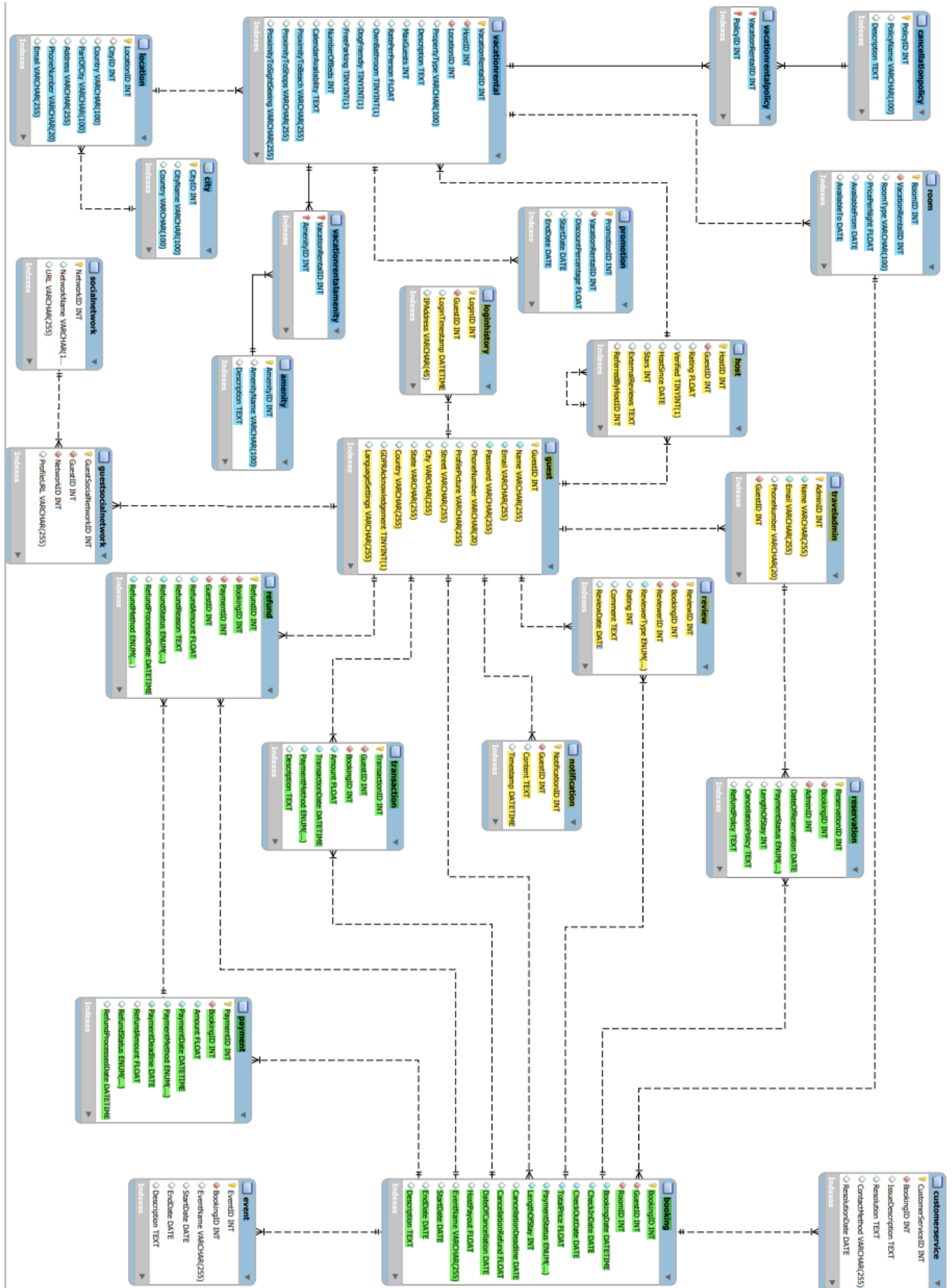
TravelAdmins: TravelAdmins oversee the platform's operations, manage user accounts, and handle customer service issues. Access to all the data stored in the database and the ability to generate reports is required for their role.

Data and Functional Requirements:

In response to feedback, the database requirements have been adjusted to better align with platform functionality. Specifically, payment processing and transactions are now connected for a more realistic representation of the payment flow, while reviews can now be left by both Guests and Hosts. The database manages data related to users, properties, bookings, payments, and reviews, ensuring secure and consistent data management for all interactions.

2. Detailed Explanation of the EER Diagram

The Enhanced Entity-Relationship (EER) diagram reflects these updated requirements, incorporating tutor feedback regarding who writes reviews and the connection between transactions and payments.



Entity and Attribute Design:

Each entity was designed to include the appropriate attributes, which were matched to the data requirements identified earlier. The key attributes were clearly marked to indicate their importance, and the relationships between tables are well-structured to handle both guest and host reviews and the payment flow.

Relationships:

The revised EER diagram now includes clearer connections between transactions, bookings, and payments, reflecting the enhanced realism in the payment process. The review entity has been updated to allow both Guests and Hosts to leave reviews, with attributes for ReviewerType and ReviewerID. Triple relationships are also included for complex interactions, and recursive relationships remain in the Host entity for tracking referrals.

Chen notation was again used to define cardinality, ensuring logical connections between entities.

Normalization:

During the design process, normalization was prioritized to minimize redundancy and enhance data integrity. The City entity centralizes city information, while full address details (e.g., street, state, country) are stored in relevant entities. These are linked through foreign keys to ensure consistent address data across the database without duplication.

3. Tools and Documentation

MySQL Workbench was used to create the updated EER diagram and manage the design process. The data dictionary has been expanded to reflect changes made during the tutor feedback phase, detailing new attributes such as ReviewerType and the integrated payment flow.

4. Data Dictionary

1. Table: Guest

Stores information about the guests using the platform.

Column Name	Data Type	Description
GuestID	INT AUTO_INCREMENT	Unique identifier for each guest (Primary Key).
Name	VARCHAR(255)	Full name of the guest.
Email	VARCHAR(255)	Email address of the guest (must be unique).
Password	VARCHAR(255)	Hashed password for the guest's account.
PhoneNumber	VARCHAR(20)	Guest's phone number.
ProfilePicture	VARCHAR(255)	URL or path to the guest's profile picture.
Address	VARCHAR(255)	Address of the guest (optional).
GDPRAcknowledgement	BOOLEAN	Indicates if the guest acknowledged GDPR terms.
LanguageSettings	VARCHAR(255)	Preferred language of the guest.

2. Table: Host

Stores information about hosts who rent out properties.

Column Name	Data Type	Description
HostID	INT AUTO_INCREMENT	Unique identifier for each host (Primary Key).
GuestID	INT	Foreign key referencing Guest (GuestID).
Rating	FLOAT	Average rating of the host.
Verified	BOOLEAN	Whether the host is verified.
HostSince	DATE	Date when the host started hosting.
Stars	INT	Number of stars or levels earned by the host.
ExternalReviews	TEXT	Reviews from external platforms.
ReferredByHostID	INT	Self-referencing foreign key to another HostID.

3. Table: TravelAdmin

Stores information about the administrators managing travel services.

Column Name	Data Type	Description
AdminID	INT AUTO_INCREMENT	Unique identifier for each admin (Primary Key).
Name	VARCHAR(255)	Full name of the administrator.
Email	VARCHAR(255)	Email address of the admin (must be unique).
PhoneNumber	VARCHAR(20)	Admin's phone number.
GuestID	INT	Foreign key referencing Guest (GuestID).

4. Table: SocialNetwork

Stores details about different social networks.

Column Name	Data Type	Description
NetworkID	INT AUTO_INCREMENT	Unique identifier for each social network.
NetworkName	VARCHAR(100)	Name of the social network (e.g., Facebook).
URL	VARCHAR(255)	URL of the social network.

5. Table: GuestSocialNetwork

Stores the connection between guests and their social network profiles.

Column Name	Data Type	Description
GuestSocialNetworkID	INT AUTO_INCREMENT	Unique identifier for each entry.
GuestID	INT	Foreign key referencing Guest (GuestID).
NetworkID	INT	Foreign key referencing SocialNetwork (NetworkID)
ProfileURL	VARCHAR(255)	URL to the guest's profile on the social network.

6. Table: LoginHistory

Stores records of guest login attempts.

Column Name	Data Type	Description
LoginID	INT AUTO_INCREMENT	Unique identifier for each login attempt.
GuestID	INT	Foreign key referencing Guest (GuestID).
LoginTimestamp	DATETIME	The date and time of the login.
IPAddress	VARCHAR(45)	IP address from which the login was made.

7. Table: Notification

Stores notifications sent to guests.

Column Name	Data Type	Description
NotificationID	INT AUTO_INCREMENT	Unique identifier for each notification.
GuestID	INT	Foreign key referencing Guest (GuestID).
Content	TEXT	The content of the notification.
Timestamp	DATETIME	Date and time the notification was sent.

8. Table: City

Stores city information.

Column Name	Data Type	Description
CityID	INT AUTO_INCREMENT	Unique identifier for each city.
CityName	VARCHAR(100)	The name of the city.
Country	VARCHAR(100)	The country where the city is located.

9. Table: Location

Stores information about specific locations for vacation rentals.

Column Name	Data Type	Description
LocationID	INT AUTO_INCREMENT	Unique identifier for each location.
CityID	INT	Foreign key referencing City (CityID).
Country	VARCHAR(100)	Country where the location is situated.
PartOfCity	VARCHAR(100)	Specific part of the city (neighborhood, etc.).
Address	VARCHAR(255)	Full address of the location.
PhoneNumber	VARCHAR(20)	Contact phone number for the location.
Email	VARCHAR(255)	Contact email for the location.

10. Table: VacationRental

Stores vacation rental property details.

Column Name	Data Type	Description
VacationRentalID	INT AUTO_INCREMENT	Unique identifier for each vacation rental.
HostID	INT	Foreign key referencing Host(HostID).
LocationID	INT	Foreign key referencing Location(LocationID).
PropertyType	VARCHAR(100)	Type of property (e.g., apartment, villa).
Description	TEXT	Description of the property.
MaxGuests	INT	Maximum number of guests allowed.
RatePerPerson	FLOAT	Rate per person per night.
OwnBathroom	BOOLEAN	Whether the property has its own bathroom.
DogFriendly	BOOLEAN	Whether pets are allowed.
FreeParking	BOOLEAN	Whether the property offers free parking.
NumberOfBeds	INT	Number of beds in the property.
CalendarAvailability	TEXT	Calendar showing available dates.

11. Table: Room

Stores individual room details within a vacation rental.

Column Name	Data Type	Description
RoomID	INT AUTO_INCREMENT	Unique identifier for each room.
VacationRentalID	INT	Foreign key referencing VacationRental (VacationRentalID).
RoomType	VARCHAR(100)	Type of room (e.g., standard, deluxe).
PricePerNight	FLOAT	Price per night for the room.
AvailableFrom	DATE	Date when the room becomes available.
AvailableTo	DATE	Date when the room is no longer available.

12. Table: Amenity

Stores details about amenities offered in vacation rentals.

Column Name	Data Type	Description
AmenityID	INT AUTO_INCREMENT	Unique identifier for each amenity.
AmenityName	VARCHAR(100)	Name of the amenity (e.g., Wi-Fi, Parking).
Description	TEXT	Description of the amenity.

13. Table: VacationRentalAmenity

Links vacation rentals with the amenities they offer.

Column Name	Data Type	Description
VacationRentalID	INT	Foreign key referencing VacationRental (VacationRentalID).
AmenityID	INT	Foreign key referencing Amenity (AmenityID).

14. Table: CancellationPolicy

Stores details about cancellation policies.

Column Name	Data Type	Description
PolicyID	INT AUTO_INCREMENT	Unique identifier for each policy.
PolicyName	VARCHAR(100)	Name of the cancellation policy (e.g., Flexible, Strict).
Description	TEXT	Description of the policy.

15. Table: VacationRentalPolicy

Links vacation rentals with their cancellation policies.

Column Name	Data Type	Description
VacationRentalID	INT	Foreign key referencing VacationRental (VacationRentalID).
PolicyID	INT	Foreign key referencing CancellationPolicy (PolicyID).

16. Table: Booking

Stores details about guest bookings.

Column Name	Data Type	Description
BookingID	INT AUTO_INCREMENT	Unique identifier for each booking.
GuestID	INT	Foreign key referencing Guest (GuestID).
RoomID	INT	Foreign key referencing Room (RoomID).
BookingDate	DATETIME	Date and time when the booking was made.
CheckInDate	DATE	Date of check-in.
CheckOutDate	DATE	Date of check-out.
TotalPrice	FLOAT	Total price for the booking.

Column Name	Data Type	Description
PaymentStatus	ENUM	Status of the booking payment. ('Paid', 'Pending', 'Cancelled')
LengthOfStay	INT	Number of nights of the stay.
CancellationDeadline	DATE	Last date to cancel for a refund.
CancellationRefund	FLOAT	Amount refunded if the booking is cancelled.

17. Table: Payment

Stores payment details related to bookings.

Column Name	Data Type	Description
PaymentID	INT AUTO_INCREMENT	Unique identifier for each payment.
BookingID	INT	Foreign key referencing Booking (BookingID).
Amount	FLOAT	Amount paid by the guest.
PaymentDate	DATETIME	Date when the payment was made.
PaymentMethod	ENUM	Method used to make the payment. ('CreditCard', 'PayPal', etc.)
Taxes	FLOAT	Applicable taxes on the booking.
CleaningCosts	FLOAT	Additional cleaning costs.
PaymentDeadline	DATE	Deadline for making the payment.
Commission	FLOAT	Commission taken from the payment.

18. Table: Reservation

Stores reservations made by administrators.

Column Name	Data Type	Description
ReservationID	INT AUTO_INCREMENT	Unique identifier for each reservation.

Column Name	Data Type	Description
BookingID	INT	Foreign key referencing Booking (BookingID).
AdminID	INT	Foreign key referencing TravelAdmin (AdminID).
DateOfReservation	DATE	Date when the reservation was made.
PaymentStatus	ENUM	Status of the reservation payment. ('Paid', 'Pending', 'Cancelled')
LengthOfStay	INT	Duration of the reservation in nights.
CancellationPolicy	TEXT	Cancellation policy terms.
RefundPolicy	TEXT	Refund policy terms.

19. Table: Review

Stores reviews left by guests after their stay.

Column Name	Data Type	Description
ReviewID	INT AUTO_INCREMENT	Unique identifier for each review. (Primary Key.)
BookingID	INT	Foreign key referencing Booking (BookingID).
ReviewerID	INT	Foreign key referencing Guest (GuestID) for guests and hosts.
ReviewerType	ENUM('Guest', 'Host')	Indicates whether the review was made by a guest or a host.
Rating	INT	Rating given by the guest (1-5).
Comment	TEXT	Guest's feedback.
ReviewDate	DATE	Date when the review was submitted.

20. Table: CustomerService

Stores customer service interaction details.

Column Name	Data Type	Description
CustomerServiceID	INT AUTO_INCREMENT	Unique identifier for each customer service entry.
BookingID	INT	Foreign key referencing Booking (BookingID).
IssueDescription	TEXT	Description of the issue reported by the guest.
Resolution	TEXT	How the issue was resolved.
ContactMethod	VARCHAR(255)	Method used for contacting customer service.
ResolutionDate	DATE	Date when the issue was resolved.

21. Table: Event

Stores event information related to bookings.

Column Name	Data Type	Description
EventID	INT AUTO_INCREMENT	Unique identifier for each event.
BookingID	INT	Foreign key referencing Booking (BookingID).
EventName	VARCHAR(255)	Name of the event (e.g., Berlin Marathon).
StartDate	DATE	Start date of the event.
EndDate	DATE	End date of the event.
Description	TEXT	Description of the event.

22. Table: Promotion

Stores promotion details for vacation rentals.

Column Name	Data Type	Description
PromotionID	INT AUTO_INCREMENT	Unique identifier for each promotion.
VacationRentalID	INT	Foreign key referencing VacationRental (VacationRentalID).
DiscountPercentage	FLOAT	Percentage discount applied during the promotion.

Column Name	Data Type	Description
StartDate	DATE	Start date of the promotion.
EndDate	DATE	End date of the promotion.

23. Table: Transaction

Stores financial transactions made by guests.

Column Name	Data Type	Description
TransactionID	INT AUTO_INCREMENT	Unique identifier for each transaction.
GuestID	INT	Foreign key referencing Guest (GuestID).
Amount	FLOAT	Amount of money transacted.
TransactionDate	DATETIME	Date and time of the transaction.
PaymentMethod	ENUM	Method used for the transaction. ('CreditCard', 'BankTransfer', 'PayPal', etc.)
Description	TEXT	Description of the transaction.

24. Table: Refund

Stores refund details for bookings.

Column Name	Data Type	Description
RefundID	INT AUTO_INCREMENT	Unique identifier for each refund.
BookingID	INT	Foreign key referencing Booking (BookingID).
PaymentID	INT	Foreign key referencing Payment (PaymentID).
GuestID	INT	Foreign key referencing Guest (GuestID).
RefundAmount	FLOAT	Amount refunded to the guest.
RefundReason	TEXT	Reason for the refund.
RefundStatus	ENUM	Status of the refund. ('Pending', 'Processed')

Column Name	Data Type	Description
RefundProcessedDate	DATETIME	Date and time when the refund was processed.
RefundMethod	ENUM	Method used for the refund. ('CreditCard', 'PayPal', 'GPay', 'BankTransfer', etc.)